From Sleepless to SleepSmart: A Sleep-Tracking Application to Promote Healthy Sleeping Habits and Reduce Sleep Deprivation

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I. Introduction

Sleep is a vital component as in day to day life as it is necessary for maintaining good physical, mental and emotional health. As sleep allows the body to rest thus supporting brain function, improving concentration and aids in regulating important body processes. However, many individiuals expereince sleep deprivation, which occurs when an individual does not get enough quality sleep or has an irregular sleeping pattern. According to the Centers for Disease Control and Prevention, about 1 in 3 adults in the United States reported not getting enough sleep or rest everyday. 40% of the adults report falling asslep during the day without meaning to at least once a month (*What Are Sleep Deprivation and Deficiency? | NHLBI, NIH*, 2022). Factors such as academic and work demands, stress, excessive screen time, unhealthy sleeping habits and use of stimulants such as caffeine contribute to this insufficient sleep. And when experienced regularly, sleep deprivation can negatively affect daily functioning, mood and overall health. With all this in mind, this paper proposes the development of a mobile application designed to help users monitor and understand their sleep habits. This application aims to provide users with personalized sleep information, reminders, and recommendations that encourage better sleep practices.

II. Problem Description

Sleep deprivation is a common problem that affects people's physical health and mental well-being, and ability to perform daily activities. Many individuals, specifically students and working adults, do not get enough sleep because of busy schedules, academic or work demands, stress, excessive use of mobile devices and inconsistent sleeping habits (*Sleep Deprivation*, 2026). But despite experiencing tiredness and the effects of inadequate sleep, some individuals may not be fully aware of how their daily routines affect the quality

and duration of their sleep. One of the major factors that affect this is due to the lack of regular monitoring and awareness of personal sleep patterns (Hanson & Huecker, 2023). Individuals often sleep whenever it's convenient for them or when the sleepiness start to take over and they are unable to stay awake. This leads to unhealthy sleeping habits, as individuals do not keep track of when they sleep, how long they sleep or if their sleeping schedule is consistent. Without this information. It can be difficult for individuals to identify unhealthy sleep habits and make appropriate changes

III. Proposal Solution

The proposed solution to reduce sleep deprivation among students and working adults is through the development of an application named SleepSmart. SleepSmart is a mobile sleep-tracking application designed to help users monitor, understand and improve their sleeping habits. The application allows users to record and monitor important sleep information such as bedtime, wake-up time, total sleep duration and sleep schedule consistency using phone sensors and manual input to estimate and record sleep patterns. The collected data will be presented through simple charts and summaries, allowing the user to easily understand their sleeping habits and identify patterns of insufficient or irregular sleep.

The application will also provide personalized reminders and recommendations to encourage healthier sleeping behaviors through notifications, and updates on maintaining a consistent sleeping schedule and to limit screen use before sleeping. Another nifty feature is the sleep-deprivation alert system. When the application detects repeated short sleep duration and irregular sleeping patterns, it notify's the user and provides practical suggestions for improving their routine. The application itself does not diagnose sleep disorders, but instead serves as a self-monitoring tool and encourages users to seek professional healthcare advice if persistent sleep problems occur.

All in all, SleepSmart, aims to make sleep monitoring convenient, accessible and easy to understand, aiding users in becoming more aware of their sleeping habits and encouraging positive changes to their sleep routines.

IV. Conclusion

Sleep deprivation is a significant problem that can negatively affect an individual's physical health, mental well being, concentration and activities of daily living. Many individuals experience inadequate sleep without fully recognizing how their habits contribute to the problem. With the rise of technology, digital tools can provide a convenient way to help individuals become more aware and manage their sleep patterns. One such way is through the app SleepSmart, which offers a practical technological solution by allowing users to monitor their sleep duration, bedtime, wake-up time and sleep consistency, through sleep trackers, reminders, summaries and personalized recommendations. Fostering the development of healthier and more consistent sleeping habits.

V. Reference

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